

# **Alkaline phosphatases as a biomarker for hepatobiliary disease**

*Case Study for the CFDE BiomarkerKB Project*

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## **Introduction:**

Alkaline phosphatases (ALPs) are a group of iso-enzymes located on the outer layer of the cell membrane (1). They catalyze the hydrolysis of organic phosphate esters found in the extra-cellular space. Zinc and magnesium are essential cofactors of this enzyme.

ALPs are found in decreasing concentration in various organs, such as the placenta, intestine (ileum), kidney, bone, and liver. ALPs are categorized into two types: tissue-specific and tissue non-specific. The tissue – nonspecific ALPs are clinically significant, as they constitute most of the circulating fraction in serum (2). They are encoded by a single gene and expressed in liver, bone, and kidney (1). Although all tissue nonspecific ALPs share an identical amino acid sequence, they possess distinct carbohydrate and lipid side chains. These unique physiochemical properties are conferred through post – translational modification (3) and can be used further to identify the organ of origin.

## **Evidence supporting the use of alkaline phosphate as a biomarker for hepatobiliary disease.**

Over 80% of the ALP in serum originates from the liver and bone, with minor contributions in some people from the intestine (4). Liver and bone ALPs can be further distinguished by their physiochemical properties, especially thermal sensitivity. Thus, the monitoring of serum ALP can be useful in identifying disorders of the hepatobiliary system and of bone.

## **Biomarker precision:**

Serum ALP levels exhibit age-related variation in healthy individuals (1). The levels are highest during childhood and puberty due to bone growth and development and then decrease as the individuals age. The decline in serum ALP levels within the age group of 15 to 50 is slightly more pronounced in men than women (5). Serum ALP levels increase again during old age with a much notable gender distribution difference. The underlying reasons for these typical fluctuations remain unidentified. Research indicates a positive correlation between body weight and smoking with a negative correlation with height (7). Normal serum ALP levels adjusted for age and sex are widely available in the pathology literature or established by individual laboratories themselves.

The principal clinical significance of measuring ALP levels is in diagnosing hepatobiliary disease. Most substantial ALP elevations are observed in patients with cholestasis. Usually, elevations four – fold or more above the upper limit of normal can be found and up to 75% of patients with cholestasis, whether intra- or extra hepatic (6). The degree of elevation does not help to distinguish between the two types of cholestasis. ALP

elevations are observed in cases of cholestasis due to obstruction due to cancer, choledocholithiasis, biliary stricture, sclerosing cholangitis. Intrahepatic elevated ALP levels can also be seen in disorders leading to intrahepatic stasis such as Primary Biliary Sclerosis (PBS), drug -induced liver injury, chronic rejection of liver allograft, infiltrative liver disease such as sarcoidosis and severe alcoholic hepatitis leading to steatonecrosis (7).

A moderate elevation in serum ALP, which can reach up to four times the upper limit of normal, is nonspecific, as it could manifest in various liver – related conditions, including cirrhosis, chronic hepatitis, viral hepatitis, ischemic, cholangiopathy, and congestive heart failure. Notably, disorders that do not primarily involve the liver, such as abdominal infections, cholestasis of sepsis, Hodgkin lymphoma, myeloid metaplasia, can also lead to a moderate elevation in serum ALP (6).

#### **How additional illumination could improve their clinical value:**

When elevated an serum ALP levels is detected, identification of its source should be determined. This information can be obtained through various additional markers. The most common is the elevation of other serum markers related to hepatobiliary disease that would support the hepatic origin. These include the serum aminotransferases, alanine transaminase (ALT) and aspartate aminotransferase (AST). Additional information can be obtained by biomarkers for hepatocellular damage such as gamma-glutamyl transferase (GGT). When ALP is the sole biochemical test that shows elevation while the other hepatobiliary enzymes remain within normal limits or when ALP is disproportionally increased compared with other liver enzymes, it is critical to pinpoint the primary source of the abnormality (6). This is accomplished using the different physiochemical properties that ALP derived from liver, bone, placenta and intestines have (1). Available methods include serum electrophoresis and a direct immunochemical method for assessment of bone-related ALP (9). Another approach utilizes the differences in heat stability in the isoenzymes (1); however, the method is imprecise.

#### **How future development could improve their clinical value:**

Within a semantic framework such as a biomarker knowledge graph, integrating multi-tissue transcriptomic, proteomic, and post-translational modification data could potentially map specific ALP glycoforms directly to distinct pathophysiological states. Also, applying some type of causal machine learning to longitudinal clinical and proteomics/genomics datasets can illuminate the exact confounding factors such as age, biological sex, body weight, and other comorbidities that naturally drive baseline ALP fluctuations. Developing these types of refined ALP signatures could help transform ALP from a rather crude, non-specific screening metric into a more sensitive predictive metric for hepatobiliary disease

**Table 1:** LOINC codes and related lab test information for Alkaline phosphatases as biomarkers

| LOINC Code | Lab Name, Details                                                                                                   | Protein                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 6768-6     | Alkaline Phosphatase [Enzymatic activity/volume] in Serum or Plasma                                                 | Alkaline Phosphatase                                                                                              |
| 1783-0     | Alkaline Phosphatase Isoenzymes [Enzymatic activity/volume] in Serum or Plasma                                      | Alkaline Phosphatase Isoenzymes                                                                                   |
| 1784-8     | Alkaline Phosphatase - bone [Enzymatic activity/volume] in Serum or Plasma                                          | Bone-Specific Alkaline Phosphatase (ALPL)                                                                         |
| 1782-2     | Alkaline Phosphatase - liver [Enzymatic activity/volume] in Serum or Plasma<br>Liver-Specific Alkaline Phosphatase  | Alkaline Phosphatase.liver [Enzymatic activity/volume] in Serum or Plasma<br>Liver-Specific Alkaline Phosphatase  |
| 1781-4     | Alkaline Phosphatase - intestinal [Enzymatic activity/volume] in Serum or Plasma<br>Intestinal Alkaline Phosphatase | Alkaline Phosphatase.intestinal [Enzymatic activity/volume] in Serum or Plasma<br>Intestinal Alkaline Phosphatase |
| 27341-7    | Alkaline Phosphatase - placental [Enzymatic activity/volume] in Serum or Plasma                                     | Placental Alkaline Phosphatase                                                                                    |
| 19023-1    | Alkaline Phosphatase - leukocyte [Enzymatic activity/volume] in Blood                                               | Leukocyte Alkaline Phosphatase (LAP)                                                                              |

## References:

1. D Lowe, T Sanvictores, M. Zubair, S John. Alkaline Phosphatase. StatPearls (internet): Treasure Island (FL): StatPearls Pub; Oct 2023 PMID: 29083622.
2. Zaher DM et al. Recent advances with alkaline phosphate, isoenzymes and their inhibitors. Arch Pharm (Weinheim). 2020 May; 353 (5): e2000011).

3. Dixon IMC. Tissue nonspecific alkaline phosphate (TNAP): a player in post – MI cardiac fibrosis. *EBioMedicine*: 2021 June; 68:103430.
4. Crofton PM. Biochemistry of alkaline phosphatase isoenzymes. *Crit Rev Clin Lab Sci*. 1982; 16 (3): 161–94.
5. Green MR, Sambrook J. Alkaline phosphatase. *Cold Spring Harb Protoc*. 2020 August 03; 2020 (8); 100768.
6. Sharma U, Pal D, Prasad R. Alkaline phosphatase: an overview. *Indian J Clin Biochem*. 2014 July; 29 (3) 269–78.
7. Brichacek AL, brown CM. Alkaline phosphatase: a potential biomarker for stroke and implications for treatment. *Metab Brain Dis*. 2019 February; 34 (1):3–19.